THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these would you NOT regard as a renewable source of water? 1

(a) fossil groundwater in deep layers **(b)** rivers
(c) lakes **(d)** rainfall

2. The Moon has extreme temperature changes because it: 1

(a) has water **(b)** has oceans
(c) has no atmosphere **(d)** has a magnetic field

3. The thickest layer of the Earth is the: 1

(a) outer core **(b)** mantle **(c)** crust **(d)** inner core

4. Temperature increases in the stratosphere because: 1

(a) ozone absorbs UV radiation **(b)** clouds
(c) greenhouse gases **(d)** the Sun is closer

5. Groundwater is depleted when: 1

(a) more is drawn than is recharged **(b)** trees are planted
(c) rainwater is harvested **(d)** it rains heavily

6. Which of these is NOT needed for life as we know it? 1

(a) water **(b)** air
(c) suitable temperature **(d)** diamonds

7. The average temperature in the Earth's core is about: 1

(a) 0 degrees Celsius **(b)** 100 degrees Celsius
(c) thousands of degrees Celsius **(d)** -50 degrees Celsius

8. A weather balloon bursts at a great height because: 1

(a) ozone is present
(b) it is too heavy
(c) it freezes
(d) the outside pressure is low and the gas inside expands

9. Rainwater harvesting increases: 1

(a) ozone **(b)** pollution
(c) temperature **(d)** groundwater level

10. The Earth's distance from the Sun is about: 1

(a) 150 million km **(b)** 1500 km **(c)** 1.5 billion km **(d)** 15 million km

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain how seismic waves help us study the interior of the Earth. 2

12. Why does the atmosphere help in keeping the Earth's temperature steady? 2

13. Why is the ozone layer more important in the stratosphere than at ground level? 2

14. How does the thick atmosphere of Venus affect its temperature? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the troposphere and the stratosphere (three points). 3

16. Explain the process of desalination and its use. 3

17. Discuss two ways in which mining and construction affect the Earth's surface and ecosystems. 3

OR

Explain how a volcano and an earthquake are related to Earth's interior. 3

18. Explain how an individual can reduce his carbon footprint. 3

OR

Compare the Earth with the Moon in terms of gravity, atmosphere and water (three points). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how the greenhouse effect operates, how human activities enhance it, and compare Earth, Venus and Mars. 5

OR

Discuss the sources, effects and control of air pollution and water pollution with two examples each. 5

20. Explain why the Earth is the only known planet with life, listing at least five conditions and comparing with other bodies. 5

OR

Explain how scientists study the Earth's past climate using ice cores, tree rings and fossils. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND